

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Sixth Semester of B. Tech. (CL) Examination

November 2012

CL307 Structural Analysis-III

Date: 06.11.2012, Tuesday

Time: 10:00 a.m. To 01:00 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

- Q - 1(a) A pre-tensioned concrete beam, 100 mm wide and 300 mm deep, is prestressed by straight wires carrying an initial force of 150 kN at an eccentricity of 50 mm. The modulus of elasticity of steel and concrete are 210 and 35 kN/mm² respectively. Estimate the percentage loss of stress in steel due to elastic deformation of concrete if the area of steel wires is 188 mm². [07]
- Q - 2 (a) A rectangular concrete beam, 100 mm wide by 250 mm deep, spanning over 8 m is prestressed by a straight cable carrying an effective prestressing force of 250 kN located at an eccentricity of 40 mm. The beam supports a live load of 1.2 kN/m. Calculate the resultant stress distribution for the central cross section of the beam. The density of concrete is 24kN/m³. [04]
- (b) Analyse the continuous beam shown in Figure 1 by moment Distribution Method, if support B sinks by 10 mm. Take $EI = 4000 \text{ kN-m}^2$. Draw Bending moment diagram. [10]

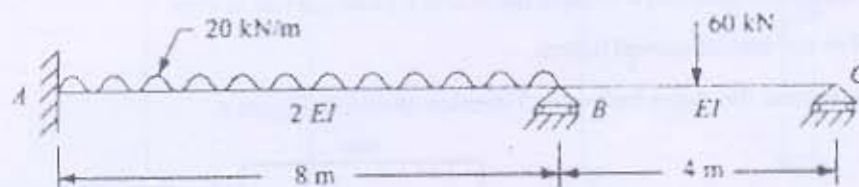


Figure 1
OR

- (b) Analyse the frame shown in Figure 2 by Moment Distribution Method. Draw Bending moment diagram. [10]

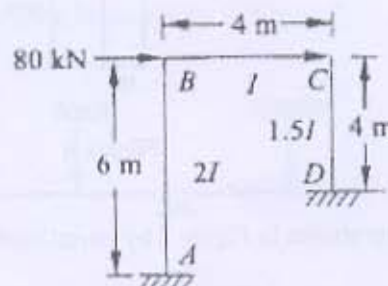


Figure 2

- Q - 3 Using slope deflection method, determine the support moments and reactions for any two continuous beams. Refer Figures 3, 4 & 5. Draw also S.F. and B.M. Diagram. [14]



Figure 3

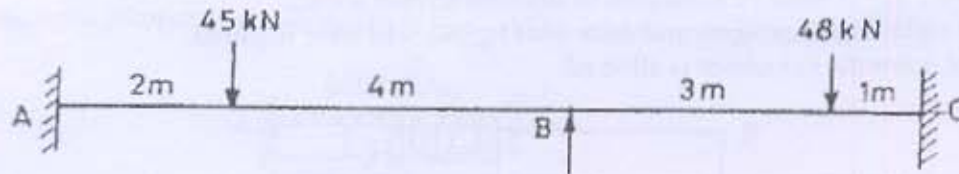


Figure 4

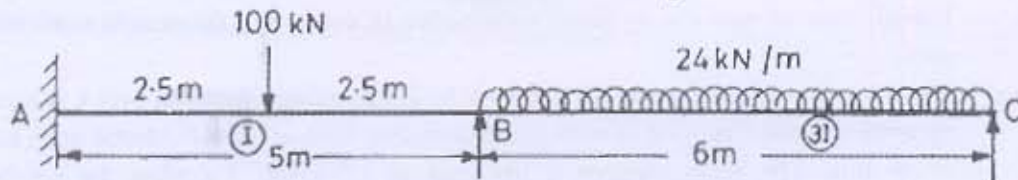


Figure 5

SECTION - II

- Q - 4(a) Distinguish between a straight beam and a beam curved in plan. [04]
 (b) Give the uses of curved beams. [03]
 Q - 5 (a) Determine the shape factor for T-section shown in Figure 6. [05]

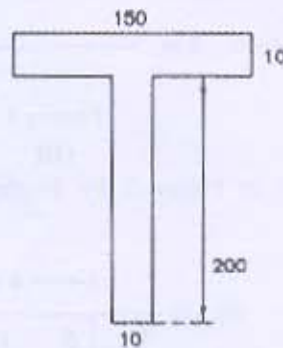


Figure 6

- (b) Attempt any One [09]
 (i) Analyse the portal frame shown in Figure 7 by portal method.

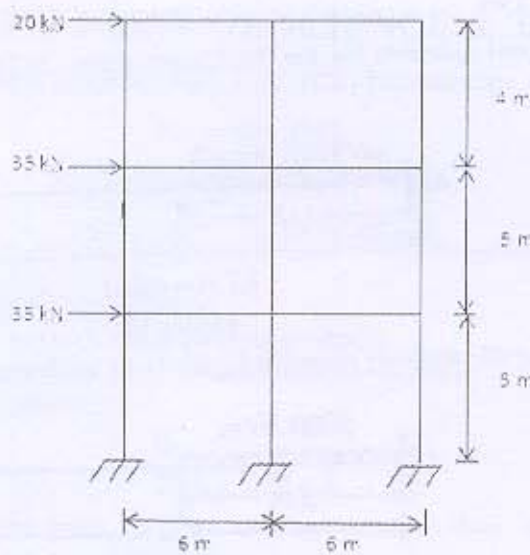


Figure 7

(ii) Analyse the frame shown in Figure 8 by cantilever method.

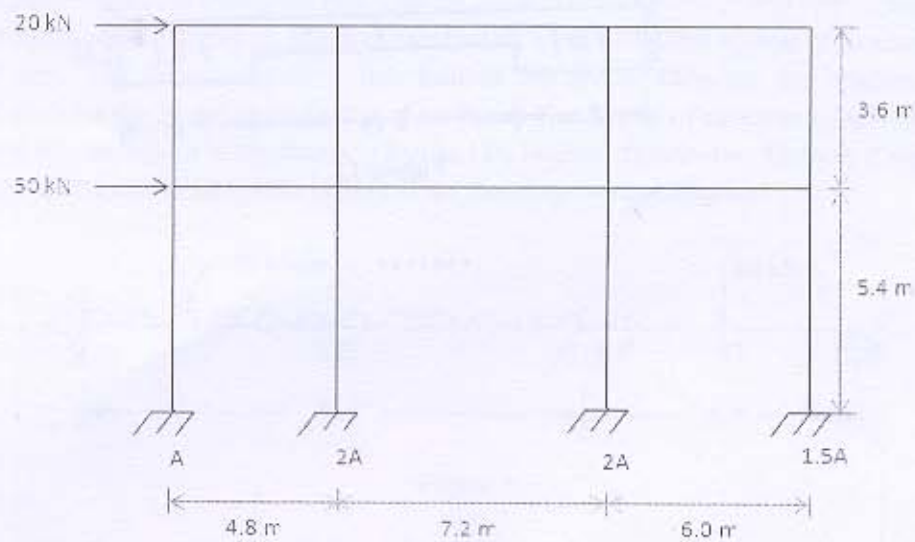


Figure 8

Q - 6 (a) Compute fully plastic moment for a beam shown in Figure 9.

[04]

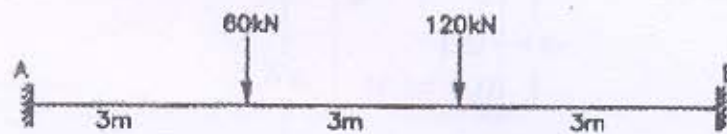


Figure 9

(b) Attempt any Two

- (i) Find the fixed end moments for the fixed beam shown in Figure 10 by method of column analogous.

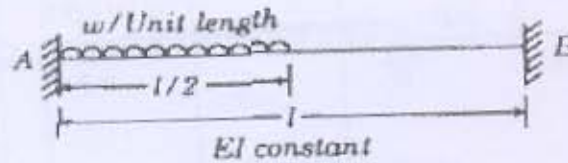


Figure 10

- (ii) Analyse the propped cantilever shown in Figure 11 by column analogy method.

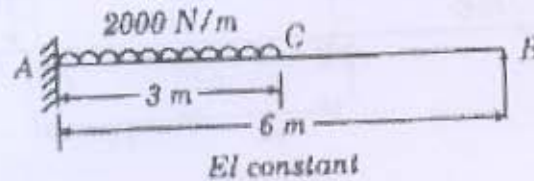


Figure 11

- (iii) Find the stiffness and carry-over factor for the fixed beam shown in Figure 12.

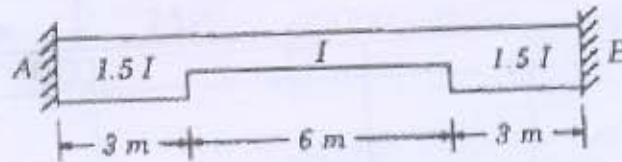


Figure 11
